



Tertiary Infotech Academy Pte Ltd

UEN: 201200696W

LESSON PLAN

For

AI Vibe Coding for React Development

Course Code: C1143

Conducted by

Tertiary Infotech Academy Pte Ltd

UEN: 201200696W

Version 4.0

DOCUMENT VERSION CONTROL RECORD

Version Number	Effective Date of Release	Summary of Included Changes	Author
1.0	10 July 2026	Initial release — 2-day lesson plan aligned to the 24 hands-on labs across six topics (vibe coding, cloud deployment, core React, hooks, Neon Postgres APIs, and React Router).	Dr. Alfred Ang
2.0	11 July 2026	Added the four standalone illustration projects to the lab reference; schedule unchanged.	Dr. Alfred Ang
3.0	11 July 2026	Restructured around ONE progressively-built full-stack project on Neon + Vercel; removed the side projects and added the mini-capstone lab (Lab 6.6 — add a course-reviews feature). 25 labs; schedule unchanged.	Dr. Alfred Ang
4.0	11 July 2026	Replaced the running project with Cook & Bake Academy (a cooking & bakery school website, deployed live). Re-architected Topic 5 from the browser-side Neon Data API + RLS to a three-tier serverless API (/api over Neon, bcrypt + JWT). Added explicit coverage of Babel/JSX compilation, the real DOM vs the Virtual DOM, and useRef; removed the JavaScript-refresher prerequisite (JavaScript is now assumed). Schedule unchanged.	Dr. Alfred Ang

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Course Information

Course Title	AI Vibe Coding for React Development
Course Code	C1143
Training Provider	Tertiary Infotech Academy Pte Ltd (UEN 201200696W)
Duration	2 days · 7.5 instructional hours per day (15 hours)
Daily Timing	9:30 am – 5:30 pm (30-minute lunch)
Mode	Instructor-led, hands-on labs in VS Code with an AI coding agent
Project	Cook & Bake Academy — a full-stack React course catalogue built across all six topics
Technology	Vite, React 19, React Router 7, Vercel serverless functions, Neon Postgres, Vercel
Trainer	Dr. Alfred Ang

Learning Outcomes

On completion of this course, learners will be able to:

- LO1: Build a React web app using vibe coding — scaffold a Vite + React project, prompt an AI coding agent, and read, understand and correct the code it generates.
- LO2: Deploy a React web app to the cloud — version the project with Git and GitHub, produce a production build, and publish it to Vercel or GitHub Pages.
- LO3: Apply core React concepts — components, JSX and how Babel compiles it, the real DOM versus the Virtual DOM, props, composition, lists and keys, conditional rendering, events and controlled forms.
- LO4: Apply React Hooks — useState, useEffect with cleanup, useRef for direct DOM access, useContext, useReducer, and custom hooks that share logic.
- LO5: Integrate a backend API and database — build serverless API routes over Neon Postgres, fetch them from React with loading and error states, and add password hashing and JWT authentication.
- LO6: Implement real application navigation with React Router — routes, nested layouts, dynamic parameters, URL state and protected routes — and deploy the full-stack app.

Learning Reinforcement

- This is a non-assessed short course: there is no written or practical assessment.
- Every lab ends with an observable 'Test it' check the learner verifies in the browser before moving on.
- The trainer checks understanding through questioning, code walk-throughs and review of each learner's running app.

Course Schedule

Day 1 — Vibe Coding, Cloud Deployment & Core React • 9:30am – 5:30pm

Time	Duration	Topic / Activity
9:30–10:00	30 min	Welcome, course introduction and ground rules
10:00–11:00	60 min	Topic 1 — Build a React Web App Using Vibe Coding: the Prompt → Generate → Read → Understand → Correct loop, JSX, components and props (concepts + demo)
11:00–12:45	105 min	Hands-on: Lab 1.1: Set Up Your Vibe Coding Workspace; Lab 1.2: Your First Vibe-Coded Component: CourseCard; Lab 1.3: Vibe-Code the Cook & Bake Academy Landing Page
12:45–13:15	30 min	Topic 2 — Deploy Your React Web App to the Cloud: Git as a safety net, the production build, CI/CD and environment variables (concepts)
13:15–13:45	30 min	Lunch break
13:45–15:00	75 min	Hands-on: Lab 2.1: Version Control with Git and GitHub; Lab 2.2: Deploy Cook & Bake Academy to Vercel; Lab 2.3: Deploy to GitHub Pages with GitHub Actions
15:00–15:45	45 min	Topic 3 — Improving Your App by Learning Core React Concepts: the Virtual DOM, composition, lists and keys, events and controlled forms (concepts + demo)
15:45–17:15	90 min	Hands-on: Lab 3.1: The Real DOM, the Virtual DOM and Babel; Lab 3.2: Components, Props and Composition; Lab 3.3: Lists, Keys and Conditional Rendering; Lab 3.4: Events, the Bakery / Cooking Filter and a Controlled Search Box
17:15–17:30	15 min	Day 1 recap and Q&A

Total instructional time: 450 minutes (7.5 hours), 9:30am – 5:30pm, with a single 30-minute lunch.

Day 2 — Hooks, Data, Routing & Capstone • 9:30am – 5:30pm

Time	Duration	Topic / Activity
9:30–9:45	15 min	Day 1 recap and knowledge check-in
9:45–10:25	40 min	Topic 4 — React Hooks (The Vibe Way): the Rules of Hooks, state as a snapshot, immutable updates and effect cleanup (concepts +

10:25–12:05	100 min	demo) Hands-on: Lab 4.1: useState — State, Snapshots and Immutability; Lab 4.2: useEffect — Side Effects and Cleanup; Lab 4.3: useRef — The DOM Escape Hatch, and What Is NOT State; Lab 4.4: useContext and useReducer — State Without Prop Drilling; Lab 4.5: Custom Hooks — Sharing Logic, Not State
12:05–12:45	40 min	Topic 5 — Giving Your App a Backend API and a Database: the three-tier model, serverless API routes over Neon, parameterised SQL, the three states of a remote read, and bcrypt + JWT authentication (concepts)
12:45–13:15	30 min	Lunch break
13:15–15:05	110 min	Hands-on: Lab 5.1: Three Tiers — A Neon Database and Your First API Route; Lab 5.2: Parameterised SQL — Dynamic Routes and an Injection Probe; Lab 5.3: Calling the API from React — Loading, Error and Success; Lab 5.4: Auth — bcrypt, JWT, and Ownership Enforced in SQL
15:05–15:30	25 min	Topic 6 — React Router for Real App Navigation: single page applications, layout routes, dynamic parameters and protected routes (concepts)
15:30–16:40	70 min	Hands-on: Lab 6.1: Routes, Links and Layout Routes; Lab 6.2: Dynamic Routes, URL State and 404s; Lab 6.3: Protected Routes and the Student Dashboard
16:40–17:15	35 min	Hands-on: Lab 6.4: Capstone — Assemble the Full-Stack App; Lab 6.5: Deploy the Full-Stack App to Vercel and Neon
17:15–17:30	15 min	Course wrap-up, next steps and Q&A

Total instructional time: 450 minutes (7.5 hours), 9:30am – 5:30pm, with a single 30-minute lunch.

Lab Reference (aligned to the course topics)

Topic	Scope	Labs
Topic 01: Build a React Web	3 labs · Day 1	Lab 1.1, Lab 1.2, Lab 1.3

App Using Vibe Coding		
Topic 02: Deploy Your React Web App to the Cloud	3 labs · Day 1	Lab 2.1, Lab 2.2, Lab 2.3
Topic 03: Improving Your App by Learning Core React Concepts	4 labs · Day 1	Lab 3.1, Lab 3.2, Lab 3.3, Lab 3.4
Topic 04: React Hooks (The Vibe Way)	5 labs · Day 2	Lab 4.1, Lab 4.2, Lab 4.3, Lab 4.4, Lab 4.5
Topic 05: Giving Your App a Backend API and a Database	4 labs · Day 2	Lab 5.1, Lab 5.2, Lab 5.3, Lab 5.4
Topic 06: React Router for Real App Navigation	6 labs · Day 2	Lab 6.1, Lab 6.2, Lab 6.3, Lab 6.4, Lab 6.5, Lab 6.6